

Data preprocessing

* File name = housing.csv

→ load .csv file to Dataframe

import pandas as pd

df = pd.read_csv("housing.csv")

→ display information of all columns

df.info()

→ display statistical information of numerical column

df.describe()

→ display the count of unique labels for "Ocean Proximity" column

df['ocean-proximity'].value_counts()

→ Display attributes having missing values count greater than 0.

df.isnull().sum()[df.isnull().sum() > 0]

Diabetes & Adult income data sets

→ Which columns in dataset had missing values?

How did you handle them?

diabetes.isnull().sum()

adult.isnull().sum()

Both datasets had no column's with missing values.

Handling: diabetes.replace('?', np.nan, inplace=True)

diabetes.dropna(inplace=True)

2) Which categorical columns did you identify in the dataset? How did you encode them?

Ans: Categorical columns identified:

GENDER
CLASS } diabetes

workclass
education
marital-status
occupation

relationship
race
gender
native-country

} adult

encoding categorical data

le = LabelEncoder()

for col in categorical_cols:

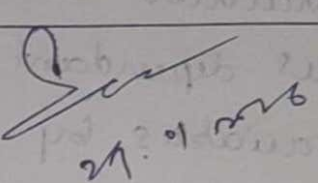
diabetes[col] = le.fit_transform(diabetes[col])

adult[col] = le.fit_transform(adult[col])

3) What is the difference between Min-max scaling & standardization? when would you use one over the other?

Min-Max Scaling	Standardization
→ Rescales features to a fixed range → $X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$ → [0, 1] range → Sensitive to outliers	→ Transforms features to have mean = 0 SD = 1 → $X_{std} = \frac{X - \mu}{\sigma}$ → centred around 0 → less sensitive

Min-max scaling is used to rescale features into fixed range (0-1) ideal for neural network or bounded data. Standardization centers features around zero with unit variance, preferred for algorithms like SVM, Regression.



Equation: $y = mx + c$
 y - predicted value
 x - input feature
 m - slope
 c - intercept

Data Set: X (features), y (target)

Import features as X
 Import target as y
 $X = X.toarray()$
 $y = y.toarray()$
 $X = X.reshape(-1, 1)$
 $y = y.reshape(-1, 1)$

Split data into training and testing sets
 $X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=0)$

Model = Linear Regression
 model.fit(X_train, y_train)
 y_pred = model.predict(X_test)

Print results
 print('R^2 score: ', r2_score(y_test, y_pred))
 print('MSE: ', mean_squared_error(y_test, y_pred))

Print coefficients
 print('Coefficients: ', model.coef_)
 print('Intercept: ', model.intercept_)

Linear Regression* Linear Regression

Is a supervised machine learning algorithm used to predict a continuous dependant variable based on one or more independent variables by fitting a best fit-straight line.

Equation : $y = mx + c$

- $y \rightarrow$ predicted value
- $x \rightarrow$ input feature
- $m \rightarrow$ slope
- $c \rightarrow$ intercept

* Data Set : House price prediction dataset.

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
np = np.sum(x)
df = pd.read_csv("data.csv")
x = df[['sqftLivArea']]
y = df['SalePrice']

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(x_train, y_train)

y_pred = model.predict(x_test)

plt.scatter(x_test, y_test, label='Actual')
plt.plot(x_test, y_pred, color='red', label='Regression Line')
plt.show()

print("MSE", mean_squared_error(y_test, y_pred))
print("R^2 score", r2_score(y_test, y_pred))
print()
```

Output :

MSE : 3009399664.466

R² Score : 0.4695955

Feature	Coefficient
GrLivArea	55.111018

Linear Regression code

import numpy as np.

x = np.array(df['x'])

y = np.array(df['y'])

m = np.sum [(x - x.mean()) * (y - y.mean())] / np.sum [(x - x.mean())²]

c = y.mean() - m * x.mean()

y_pred = m * x + c

0.5

→ Demonstrate the steps to build a machine-learning model that predicts the median housing price using the California housing price dataset.

~~Download the dataset~~

1) Perform the describe and info steps

```
housing.info()
```

```
housing.describe()
```

2) Plot the histogram of each feature. Indicate what histogram indicates on median income & how median age.

→ import matplotlib.pyplot as plt

```
housing.hist(bins=50, figsize=(15,10))
```

```
plt.show()
```

→ What histogram indicates:

i) median income

- The distribution is right-skewed (+ve)
- Most values between 286
- Few districts have income above 200
- Extreme value near 15

⇒ Since it is skewed, scaling is important before modelling.

ii) housing-median-age

- Distribution is spread between 1 and 50
- Noticeable spike at 52

⇒ Ueli effect seen.

⇒ Distribution more uniform compared to median income

3) Demonstrate pros & cons of creating test set. Write difference b/w random & stratified test set.

Random test set

- Data is split randomly into training & testing sets.
- Does not guarantee distribution of imp features.
- May cause sampling bias if dataset is skewed.
- Simple & Quick.

Stratified Test set

- Data is split while maintaining proportion of important categories.
- Ensures test set represents overall dataset properly.
- Reduces sampling bias.
- More reliable for imbalanced or skewed datasets.

4) List the geographical features from the dataset, plot graph, visualize. What does graph indicate w.r.t housing prices & location.

→ Geographical features : Latitude, Longitude.

→ What graph indicates :

- Higher prices near coastal areas.
- Lower prices in inland areas.
- Population effect (high population $\leftrightarrow$ high price regions).

5) Plot graphs to show features correlation with housing price. Which feature correlates maximum. Plot graph for flat with housing price & what it indicates.

→ Median-house-value ~~correlates maximum~~

→ Median-income v House Price

- graph shows
- clear upward trend
 - higher income $\rightarrow$ higher house price
 - Some price cap at 5,00,000.

6) list the features that could be combined to improve correlation and plot again to see if correlation has improved.

$$\rightarrow \text{total_rooms} + \text{households} = \text{rooms_per_household}$$

$$\text{total_bedroom} + \text{total_room} = \text{bedroom_per_room}$$

$$\text{population} + \text{household} = \text{pop_per_household}$$

7) list the features that need to be cleaned & demonstrated.

→ Total bedrooms

`housing_new = housing.drop("ocean_proximity", axis=1)`

`imputer.fit(housing_new)`

`housing_new_new = imputer.transform(housing_new)`

8) Is there any categorical data that needs to be converted to numerical? If so explain method used to convert & demonstrate.

→ category col: ~~ocean_proximity~~

categories: <1H ocean, inland, near ocean, near bay, island.

Method ① One-Hot-Encoding

converts each category into separate binary (0/1) nominal values column.

② label encoding.

assigns no's like inland=0, near bay=1

~~converts categorical data into numerical~~ → ordered data.

Method used ⇒ One hot encoding.

ocean proximity	inland	near bay	near ocean	<1H ocean	island
inland	1	0	0	0	0
near bay	0	1	0	0	0
near ocean	0	0	1	0	0
<1H ocean	0	0	0	1	0
island	0	0	0	0	1

1) Discuss importance of feature scaling.

Feature scaling is process of bringing all numerical features to a similar range or scale, so that no feature dominates others due to large magnitude.

Importance

- Prevents feature dominance.
- Improves model performance.
- Faster convergence.
- Required for regularization

2) Design a pipeline including - Custom transformer, feature scaler, encoder. Explain.

- Simple imputer : Fills missing values using median
- Custom transformer : creates new meaningful features, improves corr
- Standard Scaler : Scales numerical features to mean=0, std=1
- One Hot Encoder : Categorical var to binary col.
- Column transformer : Applies numerical pipeline to numeric columns
Applies ~~an~~ Encoder to categorical columns
combines all processed features to one dataset

Linear & Multi-Linear Regression

g) Solve the following Linear Regression problem using Matrix approach. Find linear regression of the data of week & product sales.

x_i (Week)	y_i (Sales)
1	2
2	4
3	5
4	9

$$Y = \begin{bmatrix} 2 \\ 4 \\ 5 \\ 9 \end{bmatrix} \quad X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

$$\beta = ((X^T X)^{-1} X^T) Y$$

$$\beta = \left(\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix}^T \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \\ 1 & 4 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \\ 9 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 4 & 10 \\ 10 & 30 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \\ 9 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 4 & 10 \\ -10 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \\ 9 \end{bmatrix}$$

$$\beta = \begin{bmatrix} -6 & -16 & -26 & -36 \\ -6 & -2 & 2 & 6 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 5 \\ 9 \end{bmatrix}$$

$$\beta = \begin{bmatrix} -530 \\ 44 \end{bmatrix}$$

$$\beta = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 6 \\ 9 \end{bmatrix} \quad \begin{matrix} 2 \times 2 \\ 2 \times 4 \\ 4 \times 1 \end{matrix}$$

$$\beta = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix} \begin{bmatrix} 20 \\ 61 \end{bmatrix} \quad \begin{matrix} 2 \times 2 \\ 2 \times 1 \end{matrix}$$

~~$$\beta = \begin{bmatrix} 1.5 & -0.5 \\ -0.5 & 0.2 \end{bmatrix}$$~~

$$\beta = \begin{bmatrix} -0.5 \\ 2.2 \end{bmatrix}$$

Canada - per capita income.csv

No major preprocessing was required.

- no missing values
- no categorical variables
- only numerical columns
- scaling was not required.

Salary.csv

Yes, preprocessing was performed.

- dropna()

hirng.csv

Yes, preprocessing was performed.

- categorical var to numerical form.
- filled NaN → median

2) Visualization on Canada Dataset

Yes, the regression line was plotted with scatterpoints.

It shows data points increasing over years, line slopes upward.

Strong +ve linear relationship b/w year & PCI

Predicted income for 2020 : 41288.69 USD

3) Hiring dataset predicted salary:

For candidate with:

12 yrs exp, 10 test score, 10 interview score

salary = 86829.13 USD

4) 1000_companies.csv - encoding & scaling

Yes, categorical variable "State" was encoded.

using One-Hot Encoding

pd.get_dummies(df, columns=['State'], drop_first=True)

Scaler was not applied.

Predicted profit = 510570.99 USD

Logistic Regression

Pass/Fail prediction of a student based on study hours.
 $a_0 = -5$ $a_1 = 0.8$

a) Write logistic regression eqn

$$Z = a_0 + a_1 x$$

$$\boxed{Z = -5 + 0.8x}$$

b) Calculate probability that student who studies for 7hrs will pass

$$Z = -5 + 0.8(7)$$

$$\boxed{Z = 0.6}$$

60.1% chance he/she passes.

c) Determine predicted class (P/F) for this student based on threshold of 0.5

$$P = \frac{1}{1 + e^{-Z}} = \frac{1}{1 + e^{-0.6}}$$

$$P = \frac{1}{1 + 0.5488}$$

$$\boxed{P \approx 0.6457} > 0.5 \text{ (Threshold)}$$

∴ Pass

Q) Consider $z = [2, 1, 0]$ for 3 classes.

Apply SoftMax function to find probability values of 3 classes.

$$z = [2, 1, 0]$$

$$P(z_i) = \frac{e^{z_i}}{\sum e^z}$$

$$e^0 = 1$$

$$e^1 = 2.718$$

$$e^2 = 1.389$$

$$\sum e^z = 11.107$$

$$P_1 = \frac{2.718}{11.107} = 0.245$$

$$P_2 = \frac{1.389}{11.107} = 0.125$$

$$P_3 = \frac{1}{11.107} = 0.090$$

SoftMax probabilities = $[0.665, 0.245, 0.090]$

1) For dataset file "HR - comma - sep.csv"

i) which variable had direct & clear impact on employee relations? why?

→ Satisfaction level : Stayed (0) → 0.66

Left (1) → 0.46

Employees who left had lower satisfaction

→ salary : From graph most emp who left belong to low salary

→ Promotion in last 5 years : Stayed → 0.026

Left → 0.005

ii) what was accuracy? Is it good?

76.75%

It is moderate.

Recall = 26%

zoo dataset

Yes preprocessing was performed.

→ removed animal name (not needed)

→ check NaN.

ii) Missing / Inconsistent values?

All columns had 0 NaN.

iii) What does confusion matrix tell?

→ Most values are diagonal

→ Very few misclassification

→ Overall accuracy: 93.56%

iv) Which class types were most misclassified? why?

→ class 6 - 1 misclassification

Classes 2 & 3 - minor confusion

Because, some animal classes share similar characteristics.

$$\frac{8}{9} = (0.888)$$

$$1/9 = (0.111)$$

$$1/9 + 1/9 = 2/9 = (0.222)$$

$$0.888 + 0.111 = 1.0$$

$$(0.888) \times 9 = 8.0$$

$$(0.111) \times 9 = 1.0$$

$$8.0 + 1.0 = 9.0$$

$$9.0 / 9 = 1.0$$

Decision tree

Q)

instance	a2	a3	classification.
1	Hot	High	N
2	Hot	High	N
6	Cool	H	N
7	Hot	H	N
8	Hot	Normal	Yes

$$\text{Entropy}(S) = -p_1 \log_2 p_1 - p_2 \log_2 p_2$$

$$= -\frac{1}{5} \log_2 \left(\frac{1}{5}\right) - \frac{4}{5} \log_2 \left(\frac{4}{5}\right)$$

$$= (0.2)(-2.32) - (0.8)(-0.32)$$

$$= 0.664 + 0.256$$

$$\boxed{S = 0.72}$$

~~Q2~~ a2 : Hot $\rightarrow 4$ (3 No 1 Yes)
Cool $\rightarrow 1$ (1 No)

$$P(\text{No}) = \frac{3}{4}$$

$$P(\text{Yes}) = \frac{1}{4}$$

$$\text{Entropy}(\text{Hot}) = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4}$$

$$= 0.311 + 0.5$$

$$= 0.811$$

$$\text{Entropy}(\text{Cool}) = 0 \quad (\text{all are No}).$$

$$E(a_2) = \frac{4}{5} (0.811) + \frac{1}{5} (0)$$

$$= \boxed{0.6488}$$

$$\text{Info Gain}(a_2) = 0.72 - 0.6488$$

$$= \boxed{0.071}$$

a_3 : High $\rightarrow 4$ (4 No)

Normal $\rightarrow 1$ (1 Yes)

Entropy (High) = 0

Entropy (Normal) = 0

$$E(a_3) = 0$$

$$IG(a_3) = 0.72 - 0$$

$$IG(a_3) = 0.72$$

$\therefore$ splitting node should be a_3 ($IG(a_3) > IG(a_2)$)

9) IRIS Dataset

$\rightarrow$ Accuracy = 1.0 (100%)

$\rightarrow$ Confusion matrix

all predictions are correctly classified.

off diagonal val are zero.

$\rightarrow$ No misclassification

$\rightarrow$ No classes confused.

10) Petrol consumption Dataset

$\rightarrow$ Regression tree:

splits dataset based feature values.

each leaf node predicts numerical value.

$\rightarrow$ Most imp features?

1. Avg income.

2. Petrol tax.

$\rightarrow$ Error metrics:

MAE = 94.3

RMSE = 131.71

KNN classification

Q) Consider given dataset for $k=3$ & test data $(X, 35, 100)$ as $(Person, Age, Salary, k)$ solve using KNN classification model & predict the target.

Person	Age	Salary	Target
A	18	50	N
B	23	55	N
C	24	70	N
D	41	60	Y
E	43	70	Y
F	38	40	Y
X	35	100	?

Solu : $Age_1 = 35, Salary_1 = 100 \rightarrow X$

$$Dist = \sqrt{(Age_1 - Age_2)^2 + (Salary_1 - Salary_2)^2}$$

Person	Age	Salary	Target	Dist
A	18	50	N	$\sqrt{(35-18)^2 + (100-50)^2} = 52.81$
B	23	55	N	$\sqrt{(35-23)^2 + (100-55)^2} = 46.57$
C	24	70	N	$\sqrt{(35-24)^2 + (100-70)^2} = 31.75$ ✓
D	41	60	Y	$\sqrt{(35-41)^2 + (100-60)^2} = 40.45$ ✓
E	43	70	Y	$\sqrt{(35-43)^2 + (100-70)^2} = 31.05$ ✓
F	38	40	Y	$\sqrt{(35-38)^2 + (100-40)^2} = 60.008$
X	35	100	?	

3 nearest neighbors : E, C, D

↓ ↓ ↓
Y N Y

Majority voting : Y

∴ Target $(X, 35, 100) = \underline{\underline{Y}}$

1) For Iris Dataset

How to choose the k value? Demonstrate using accuracy rate & error rate.

In KNN, the value of k (no. of neighbors) determines how the model classifies a new data point.

Method to choose k

Optimal k is chosen by testing multiple k values calculating Accuracy rate, error rate

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{Total predictions}}$$

$$\text{Error rate} = 1 - \text{Accuracy}$$

example

$$\text{if accuracy} = 1.0$$

$$\text{error rate} = 0$$

$\Rightarrow$ model predicted all test samples properly

2) Diabetes Dataset

What is the purpose of feature scaling? How to perform it?

Feature scaling is used to normalize the range of independent variables so that all features contribute equally to the model. This is important for KNN because it uses distance calculations.

Feature scaling is performed by :

$\rightarrow$ Min-Max scaling : Rescales data to fixed range (0 to 1)

$$X_{\text{scaled}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

$\rightarrow$ Standardization (z-score) : Transforms mean (μ) of 0 & SD of 1.
to have

$$X_{scaled} = \frac{x - \mu}{\sigma}$$

1) for this data set

How to choose the k value? Recommendation using cross-validation

1) for this data set

In KNN, the value of k (no. of neighbors) determines

how the model behaves a new data point

How to choose k

Optimal k is chosen by testing multiple k values and looking at the error rate

Amount of error rate

Amount of error rate = 1 - Accuracy

Amount of error rate = 1 - Accuracy

Total predictions

Example:

if error rate = 10

error rate = 0

→ more predictions all the

sample properly

2) Distance based

What is the purpose of feature scaling? then to perform it?

Feature scaling is used to normalize the range of

independent variables so that all features contribute equally

to the model. This is important for KNN because it uses

distance calculations

Feature scaling is performed by

→ min-max scaling: Rescales data to fixed range (0 to 1)

$$X_{scaled} = \frac{x - x_{min}}{x_{max} - x_{min}}$$

→ standardization (z score): Transformation (M) of x to 0 of 1

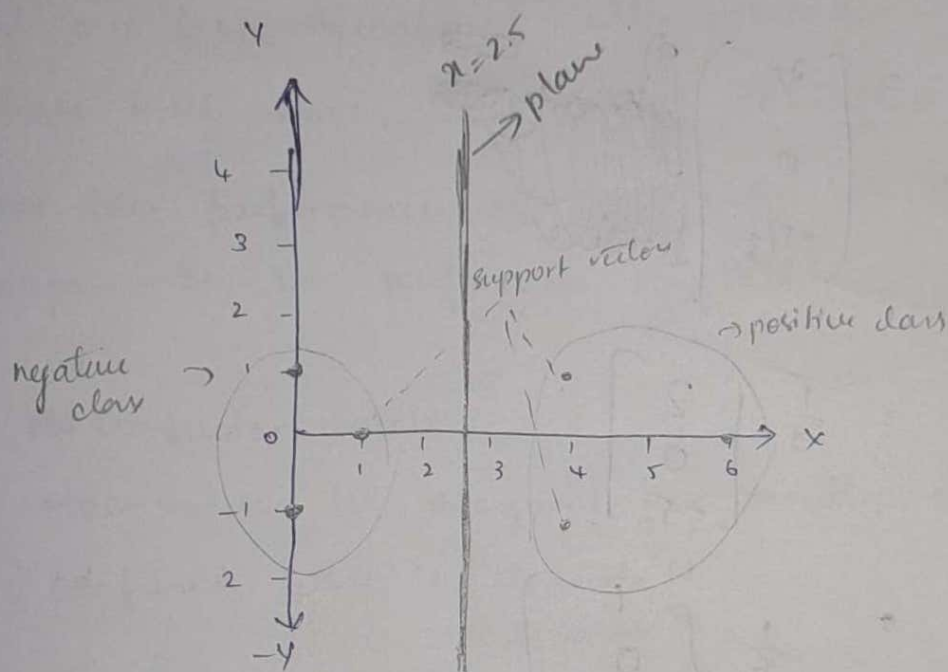
Lab - 16

~~Implement Random Forest ensemble method on a given dataset.~~

* Build Support Vector Machine model for a given dataset

Q) Points $(4,1)$, $(4,-1)$ and $(6,0)$ belong to positive class & points $(1,0)$, $(0,1)$ and $(0,-1)$ belong to negative class.

Draw an optimal hyperplane.



support vectors : $S_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $S_2 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ $S_3 = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$

$S_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ $S_2 = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$ $S_3 = \begin{pmatrix} 4 \\ -1 \end{pmatrix}$

$\alpha_1 S_1 \cdot S_1 + \alpha_2 S_2 \cdot S_1 + \alpha_3 S_3 \cdot S_1 = -1$

$2\alpha_1 + 18\alpha_2 + 5\alpha_3 = -1$ — (1)

$\alpha_1 S_1 \cdot S_2 + \alpha_2 S_2 \cdot S_2 + \alpha_3 S_3 \cdot S_2 = +1$

$5\alpha_1 + 18\alpha_2 + 16\alpha_3 = 1$ — (2)

$\alpha_1 S_1 \cdot S_3 + \alpha_2 S_2 \cdot S_3 + \alpha_3 S_3 \cdot S_3 = +1$

$5\alpha_1 + 16\alpha_2 + 18\alpha_3 = +1$ — (3)

$\alpha_1 = -22/9$

$\alpha_2 = 7/18$

$\alpha_3 = 7/18$

$$\bar{w} = \sum_{i=1}^n x_i S_i$$

$$\bar{w} = x_1 S_1 + x_2 S_2 + x_3 S_3$$

$$\bar{w} = -\frac{22}{9} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \frac{7}{18} \begin{pmatrix} 4 \\ 1 \\ 1 \end{pmatrix} + \frac{7}{18} \begin{pmatrix} 4 \\ -1 \\ 1 \end{pmatrix}$$

$$\bar{w} = \begin{pmatrix} -22/9 \\ 0 \\ -22/9 \end{pmatrix} + \begin{pmatrix} 28/18 \\ 7/18 \\ 7/18 \end{pmatrix} + \begin{pmatrix} 28/18 \\ -7/18 \\ 7/18 \end{pmatrix}$$

$$\bar{w} = \begin{pmatrix} 2/3 \\ 0 \\ -5/3 \end{pmatrix} \begin{matrix} \text{orientation} \\ \text{lie} \end{matrix}$$

$$= \frac{1}{3} \begin{bmatrix} 2 \\ 0 \\ -5 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 1 \\ 0 \\ -2.5 \end{bmatrix}$$

$$b + \frac{5}{3} = 0$$

$$= \frac{1}{6} \begin{bmatrix} 1 \\ 0 \\ -2.5 \end{bmatrix} \begin{matrix} \text{orientation} \\ \text{lie} \end{matrix}$$

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \text{ orientation} \Rightarrow \text{vertical}$$

$$-2.5 = 0$$

$$\boxed{-2.5}$$

optimal

$\therefore$ Hypoplane is $\boxed{\eta = 2.5}$ vertical plane

Q) For "iris.csv"

What is accuracy score of classifier using linear kernel & RBF kernel? Which kernel gave better performance? why?

→ 100% for both.

Both performed equally well

Reason: dataset is simple & almost linearly separable.

Q) For letter-recognition.csv

Present & interpret confusion matrix. Are there specific letters that are frequently confused with others?

What AUC score, how does it reflect model performance?

How does performance of SVM model on this dataset compare to its performance on IRIS?

→ Confusion matrix:

Most values lie diagonal ⇒ correctly classified.

confused letters: H with N
I with J
S with Z
O with Q

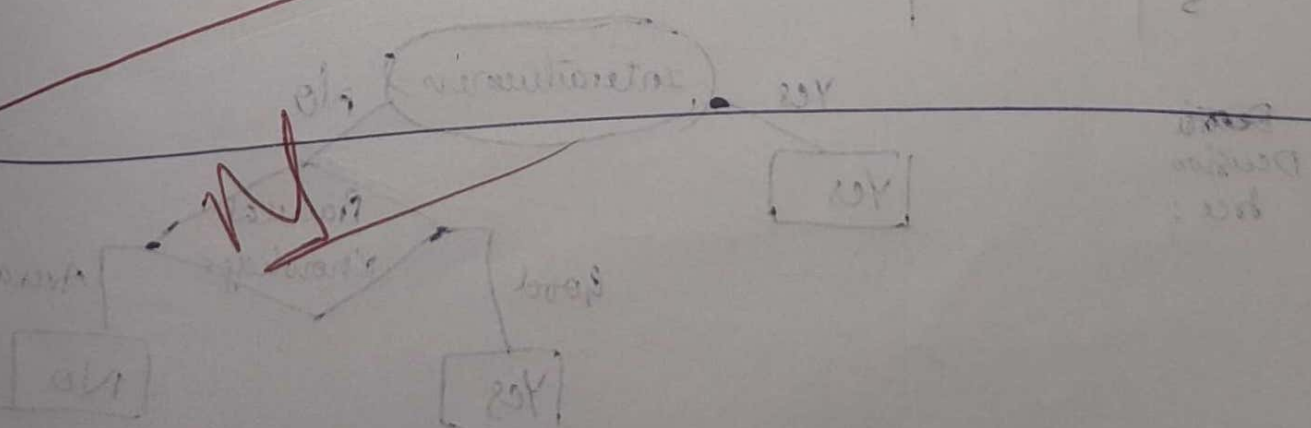
→ AUC score = 100

High classifier performance.

→ Comparison with IRIS

Accuracy (IRIS) = 100%

Accuracy (letter) = 85.45%

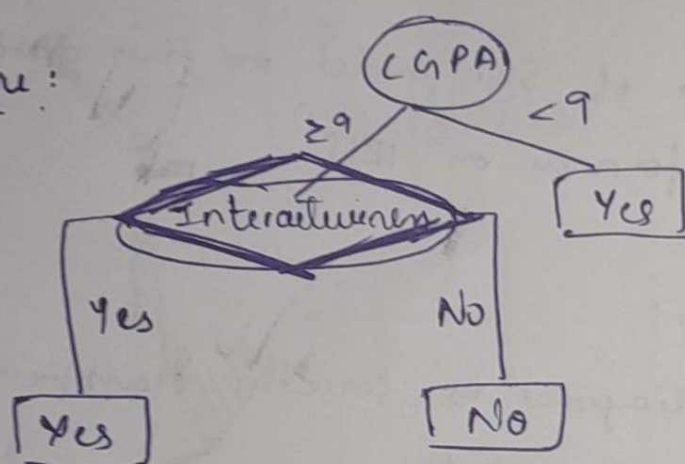


* Implement Random Forest ensemble method on given dataset.

Q) For S1 shown, draw the decision tree considering CGPA as a root node.

S.no	CGPA	interactiveness	Comm skills	Practical knowledge	Job offer
1	≥ 9	Yes	Good	Good	Yes
2	< 9	No	Moderate	Good	No Yes
3	≥ 9	No	Moderate	Avg	No
4	≥ 9	No	Moderate	Avg	No
5	≥ 9	Yes	Moderate	Good	Yes

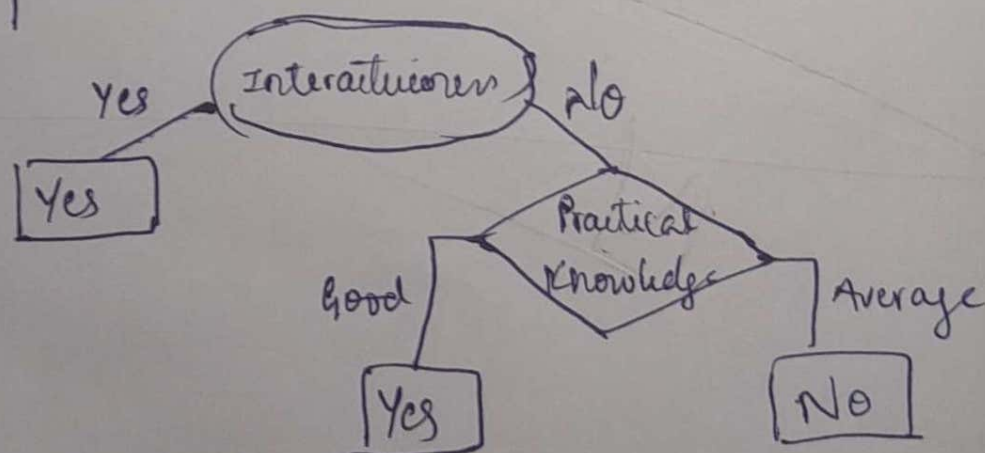
Decision tree:



Q) For sample S2 shown, draw the decision tree considering interactiveness as root node.

Sr no	CGPA	interactiveness	Comm skills	Prac. Knowledge	Job offer
1	< 9	No	Moderate	Good	Yes
2	≥ 9	No	Moderate	Avg	No
3	≥ 9	No	Moderate	Avg	No
4	≥ 9	Yes	Moderate	Good	Yes
5	≥ 9	Yes	Moderate	Good	Yes

Decision tree:



Iris dataset:

Q) What is the best accuracy score and confusion matrix of the classifier you observed & using how many trees?

→ Best accuracy obtained is 1.0 (100%)

Accuracy with 10 trees = 0.9667 (96.67%)

Best accuracy achieved using 1 tree.

Confusion matrix (for best model):

$$\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 12 \end{bmatrix}$$

(for default-10 trees):

$$\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 13 \end{bmatrix}$$

	Actual	Predicted	Count
Setosa	Setosa	Setosa	9
Setosa	Setosa	Versicolour	0
Setosa	Setosa	Virginica	0
Versicolour	Setosa	Setosa	0
Versicolour	Setosa	Versicolour	9
Versicolour	Setosa	Virginica	0
Versicolour	Versicolour	Setosa	0
Versicolour	Versicolour	Versicolour	9
Versicolour	Versicolour	Virginica	0
Versicolour	Virginica	Setosa	0
Versicolour	Virginica	Versicolour	0
Versicolour	Virginica	Virginica	12

Accuracy = $\frac{9+9+12}{36} = 1.0$

Confusion matrix for 10 trees:

$$\begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 13 \end{bmatrix}$$

$$FSE = \frac{1}{n} \sum_{i=1}^n \left(\frac{1}{2} \right) (1) + \frac{1}{2} \left(\frac{1}{2} \right) (0) = 0.5$$

$$FSE = \frac{(9+9+12)}{36} \times \frac{1}{2} = \frac{(0.5+0.5+0.5)}{36} \times \frac{1}{2} = 0.0278$$

$$(FSE = 0.5 \times 0.5 \times 0.5) + (0.5 \times 0.5) = 0.5$$

Implement Adaboosting ensemble method:

Q) Consider Adaboost algorithm. For the following sample data, show the Decision stump calculation steps for the attribute CGPA.

CGPA	Internships	Practical knowledge	Comm skills	Job
≥ 9	Yes	Good	Good	Yes
< 9	No	Good Good	moderate	Yes
≥ 9	No	Avg	moderate	No
< 9	No	Good Avg	good	No
≥ 9	Yes	Good	moderate	Yes
≥ 9	Yes	Good	moderate	Yes

Step ① random bootstrap sample:

CGPA $\geq 9 \rightarrow$ Yes.
 $< 9 \rightarrow$ No.

Step ②
$$\epsilon_i = \sum_{j=1}^N H_i(d_j) \text{wt}(d_j)$$

where $\begin{cases} H_i(d_j) = 0 & \text{if prediction correct with } H_i \\ H_i(d_j) = 1 & \text{prediction wrong.} \end{cases}$

CGPA	predicted job offer	Actual job offer	Weight
≥ 9	Yes	Yes	$1/6$
< 9	No	Yes	$1/6$
≥ 9	Yes	No	$1/6$
< 9	No	No	$1/6$
≥ 9	Yes	Yes	$1/6$
≥ 9	Yes	Yes	$1/6$

$$\epsilon_{\text{CGPA}} = \left(\frac{1}{2}\right)\left(\frac{1}{6}\right) + (1)\left(\frac{1}{6}\right) = \frac{1}{3} = 0.333$$

Step ③ calculate weight of weak classifier

$$\alpha_{\text{CGPA}} = \frac{1}{2} \frac{\ln(1 - \epsilon_{\text{CGPA}})}{\epsilon_{\text{CGPA}}} = \frac{1}{2} \times \frac{\ln(1 - 0.333)}{0.333} = 0.347$$

Step ④ Calculate normalizing factor

$$Z_{\text{CGPA}} = \left(\frac{1}{6} \times 4\right) + \left(\frac{1}{6} \times 2 \times e^{0.347}\right) = 0.9428$$

Step 2 update weights

$$wt(d_j)_{i+1} = \frac{wt(d_j)_{C_{GPA}} \text{ of correct instance} \times e^{-Z_{C_{GPA}}}}{Z_{C_{GPA}}}$$

$$= \frac{(1/6) e^{-0.347}}{0.9428}$$

$$= \underline{0.1249} \leftarrow \text{correct instances}$$

$$wt(d_j)_{i+1} = \frac{wt(d_j)_{C_{GPA}} \text{ of incorrect instance} \times e^{Z_{C_{GPA}}}}{Z_{C_{GPA}}}$$

$$= \frac{(1/6) e^{0.347}}{0.9428}$$

$$= \underline{0.2501} \leftarrow \text{incorrect instances}$$

C _{GPA}	predicted	actual	weight
Y=9	Yes	Yes	0.1249
C=9	No	Yes	0.2501
E=9	Yes	No	0.2501
Y=9	Yes	No	0.1249
C=9	No	No	0.1249
Y=7	Yes	Yes	0.1249
Y=9	Yes	Yes	0.1249

* intrain. CSV:

What is the best accuracy score & confusion matrix of classifier & using how many trees?

→ Accuracy with 10 estimators: 81.82%

Best accuracy ⇒ 83.40% → 73 estimators

Confusion matrix

(10) $\begin{bmatrix} 6782 & 632 \\ 1146 & 1211 \end{bmatrix}$

Confusion matrix

(73) $\begin{bmatrix} 7015 & 399 \\ 1223 & 1132 \end{bmatrix}$

K-means

Q) Cluster the following into 3 clusters.

$A_1(2,10), A_2(2,5), A_3(8,4), A_4(5,8)$

$A_5(7,5), A_6(6,6), A_7(1,2), A_8(4,9)$

Initial cluster centres are : $A_1(2,10)$,
 $A_4(5,8)$
 $A_7(1,2)$

Solu : iteration 1 :

points	(2,10) c1	(5,8) c2	(1,2) c3	cluster
(2,10)	0	5	9	c1
(2,5)	5	8	4	c3
(8,4)	12	4	9	c2
(5,8)	5	0	10	c2
(7,5)	10	5	9	c2
(6,6)	10	5	7	c2
(1,2)	9	10	0	c3
(4,9)	3	2	10	c2

new centres :

c1 : $A_1(2,10)$

$$c2 : \left(\frac{8+5+7+6+6}{5}, \frac{4+8+5+4+9}{5} \right) = (6,6)$$

$$c3 : \left(\frac{2+1}{2}, \frac{5+2}{2} \right) = (1.5, 3.5)$$

~~Iteration 2~~

~~(1,2) A1(2,10)~~

Implementation: Iris CSV

silhouette score : 0.47872

~~Cluster centres~~

PCA

* Implement dimensionality reduction using PCA method.

Q) In given table, reduce dimension from 2 to 1 using PCA. Compute 1st Principal component.

Feature	Example 1	Example 2	Example 3	Example 4
x_1	4	8	13	7
x_2	11	4	5	16

Given: Eigen values:

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

Eigen vectors:

$$e_1 = \begin{bmatrix} 0.5546 \\ -0.8303 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5546 \end{bmatrix}$$

Soln:

Calculate mean:

$$\bar{x}_1 = \frac{4+8+13+7}{4} = 8$$

$$\bar{x}_2 = \frac{11+4+5+16}{4} = 8.5$$

Covariance matrix:

$$\text{Cov}(x_1, x_1) = \frac{1}{N-1} \sum_{k=1}^N (x_{1k} - \bar{x}_1)(x_{1k} - \bar{x}_1)$$

$$= \frac{1}{3} ((4-8)^2 + (8-8)^2 + (13-8)^2 + (7-8)^2)$$

$$= 14$$

$$\text{Cov}(x_1, x_2) = \frac{1}{N-1} \sum_{k=1}^N (x_{1k} - \bar{x}_1)(x_{2k} - \bar{x}_2)$$

$$= \frac{1}{3} ((4-8)(11-8.5) + (8-8)(4-8.5) + (13-8)(5-8.5) + (7-8)(16-8.5))$$

$$\text{Cov}(x_2, x_1) = -11$$

$$\text{Cov}(x_2, x_2) = \frac{1}{3} ((11-8.5)^2 + (4-8.5)^2 + (5-8.5)^2 + (16-8.5)^2)$$

$$= 23$$

$$S = \begin{bmatrix} \text{Cov}(x_1, x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_2, x_1) & \text{Cov}(x_2, x_2) \end{bmatrix} = \begin{bmatrix} 14 & -11 \\ -11 & 23 \end{bmatrix}$$

eigen values :

$$\lambda_1 = 30.3849$$

$$\lambda_2 = 6.6151$$

eigen vector :

$$e_1 = \begin{bmatrix} 0.5574 \\ -0.8303 \end{bmatrix}$$

$$e_2 = \begin{bmatrix} 0.8303 \\ 0.5574 \end{bmatrix}$$

first principal ~~comp~~ component :

$$e_1^T \begin{bmatrix} x_{1k} - \bar{x}_1 \\ x_{2k} - \bar{x}_2 \end{bmatrix} = [0.5574 \quad -0.8303] \begin{bmatrix} x_{11} - \bar{x}_1 \\ x_{21} - \bar{x}_2 \end{bmatrix}$$

$$= 0.5574(4-8) - 0.8303(11-8.5)$$

$$= -4.3052$$

Feature	Ex 1	Ex 2	Ex 3	Ex 4
x_1	4	8	13	7
x_2	11	4	5	16
First principal component	-4.3052	3.7361	5.6929	-5.1238

Heart.csv

Report accuracy scores before & after applying PCA

Before PCA :

SVM : 86.4%

Logistic regression : 85.32%

Random forest : 87.5%

Best accuracy : 87.5%

After PCA

SVM : 85.86%

LR : 85.326%

Random forest : 85.87%

Best accuracy : 85.87%